

Supported Iron Nanoparticles as Catalysts for Sustainable Production of Lower Olefins

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Lower olefins are key building blocks for the manufacture of plastics, cosmetics, and drugs. Traditionally, olefins with two to four carbons are produced by steam cracking of crude oil–derived naphtha, but there is a pressing need for alternative feedstocks and processes in view of supply limitations and of environmental issues. Although the Fischer-Tropsch synthesis has long offered a means to convert coal, biomass, and natural gas into hydrocarbon derivatives through the intermediacy of synthesis gas (a mixture of molecular hydrogen and carbon monoxide), selectivity toward lower olefins tends to be low. We report on the conversion of synthesis gas to C_2 through C_4 olefins with selectivity up to 60 weight percent, using catalysts that constitute iron nanoparticles (promoted by sulfur plus sodium) homogeneously dispersed on weakly interactive α -alumina or carbon nanofiber supports.

Lower olefins (C_2 to C_4) are extensively used in the chemical industry as building blocks to synthesize a wide range of products such as polymers, solvents, drugs, cosmetics, and detergents. Traditionally, lower olefins have been produced by thermal or catalytic cracking of naphtha or vacuum gas oil (1) or from dehydrogenation of alkanes (2, 3), but environmental and economic factors are currently spurring exploration of alternative routes for their production.

In recent years there has been growing interest in the development of biomass as a renewable feedstock for the production of commodity compounds (4, 5). Pyrolyzed biomass or bio-oil can be converted catalytically to lower olefins with moderate selectivity (43% C) (5), although other compounds such as aromatics are also produced. Schemes put forward to produce lower olefins from synthesis gas (syngas)—a mixture of H_2 and CO obtained through biomass gasification—

consist of at least two conversion steps, which involve either cracking of Fischer-Tropsch (FT)–derived hydrocarbons (6) or the methanol to olefins (MTO) process (7). Here, we consider Fischer-Tropsch to olefins (FTO) as a direct route, without intermediate steps, to transform syngas into light olefins.

For several decades, research groups have attempted to develop iron-based catalysts to direct product selectivity of the FT synthesis toward light olefins (8, 9). Relative to other FT catalysts such as cobalt, iron disfavors competing formation of methane, and furthermore catalyzes the water-gas shift reaction, enabling the use of a CO-rich syngas feed without an H_2 /CO ratio adjustment. Mainly unsupported (sometimes referred to as bulk) iron oxide catalysts have been investigated (9–12), and in some cases they have exhibited high selectivities toward lower olefins (up to 70 wt %) when the iron was modified by the addition of promoters (9). Despite these promising results, however, the bulk iron catalysts are mechanically unstable when the reaction is performed at high temperature, which is necessary to steer product selectivity to lighter hydrocarbons. Under these conditions, the undesirable Boudouard reaction, $2CO(g) \rightarrow C(s) + CO_2(g)$ (13), leads to the deposition of carbon, which

can block the active sites and induce fragmentation of the particles in bulk iron catalysts (14). The poor mechanical stability of the bulk iron oxide catalysts may lead to plugging of the catalyst bed in fixed-bed operation or to fouling of separation equipment in a fluidized-bed process.

Supported iron catalysts display enhanced dispersion of the active phase and may withstand the mechanical degradation that threatens bulk iron catalysts. Research on supported iron catalysts (15–21) has met with limited success, however. Barrault *et al.* (15) found that iron dispersed on alumina with high surface area displayed much lower activity than did iron dispersed on alumina with low surface area. This finding points to a key aspect of supported iron catalysts—that is, their cumbersome activation. If highly dispersed iron oxide interacts strongly with an oxidic support with high surface area, the conversion of iron oxide into the active phase (iron carbide) is impeded (14). Other supports for iron-based FTO catalysts in addition to alumina (15, 16) have been explored, such as zeolites (18), aluminophosphate molecular sieves (19), and carbonaceous materials (20, 21). Table S1 summarizes the most relevant results reported in the literature regarding the development of carbon- or alumina-supported iron catalysts for the selective production of lower olefins. Iron supported on activated carbon displayed a high catalytic activity, but this was accompanied by low selectivity to light olefins (20) or a high deactivation rate (21). Many years of research have shown that supported iron catalysts display an inverse relationship between activity and selectivity (15).

To overcome the low activity and mechanical stability problems, we explored the use of support materials weakly interactive toward iron. As a working hypothesis, we posited that these inert supports would impart mechanical stability to the iron nanoparticles without inhibiting their activation. In particular, nanostructured carbon materials (22), such as carbon nanofibers (CNF) (23, 24) and carbon nanotubes (25), boast high specific surface area, chemical inertness, and good mechanical strength.

In addition to CNF, we explored β -silicon carbide (β -SiC) and α -alumina (α - Al_2O_3) as supports. For comparison, we also examined three bulk

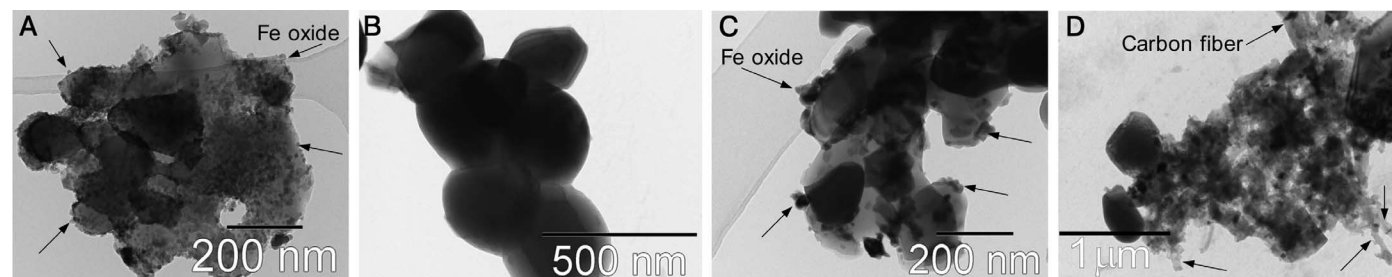


Fig. 1. TEM images of fresh and spent Fe catalysts. (A and B) The images from the fresh $Fe/\alpha-Al_2O_3$ catalysts (A) show a homogeneous distribution of iron oxide nanoparticles on the support, whereas the bulk $Fe-Ti-Zn-K$ catalyst (B) is mainly composed of aggregates of iron oxide crystals. (C and D) In

images of the spent catalysts after 64 hours of reaction at $340^\circ C$, 20 bar, and a H_2/CO ratio of 1, the $Fe/\alpha-Al_2O_3$ (C) showed sintering of Fe particles after reaction; the bulk spent catalyst (D) fragmented and manifested carbon fiber growth (indicated by arrows).

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iron catalysts (one unpromoted and two promoted) and iron supported on conventional high-surface-area SiO_2 and $\gamma\text{-Al}_2\text{O}_3$. The supported Fe catalysts were prepared using ammonium iron citrate as precursor with a nominal iron loading of 10 wt %, whereas the bulk catalysts had an iron content higher than 30 wt % (table S2). The ammonium iron citrate used in the preparation of the supported samples contained low amounts of sulfur and sodium and efficiently introduced these promoters in the catalysts (table S3).

The use of ammonium iron citrate as the metal precursor provides a homogeneous distribution of the iron nanoparticles on the support, in contrast to the extensive clustering that is observed when using iron nitrate (26). Fe nanoparticle aggregation could lead to low catalytic activity and high methane selectivity, as observed when using bulk iron catalysts. Transmission electron microscopy (TEM) was used to determine the size of the iron oxide particles and their distribution on the support.

Figure 1A shows a representative TEM micrograph of the calcined $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$, which exhibited a homogeneous distribution of iron oxide particles (arrows). The Fe_2O_3 particle size distribution (fig. S2) was 14 ± 5 nm on this support and 5 ± 1 nm on CNF. The bulk promoted catalyst

(Fe-Ti-Zn-K) consisted of large Fe_2O_3 particles (average size 400 nm), which formed aggregates resembling grape bunches (Fig. 1B).

The volume-averaged Fe_2O_3 crystallite size of the Fe catalyst precursors was calculated with the Scherrer equation using the parameters obtained by x-ray diffraction analysis (fig. S3 and table S2). Fe/SiO_2 , $\text{Fe}/\gamma\text{-Al}_2\text{O}_3$, and $\text{Fe}/\beta\text{-SiC}$ did not show the characteristic diffraction lines from iron oxide, indicating that the Fe_2O_3 was amorphous or that the crystallites were smaller than 4 nm (fig. S3).

The fresh catalysts were also analyzed by Mössbauer spectroscopy to determine the composition of the iron phase (tables S4 and S5). Iron was present in the form of hematite ($\alpha\text{-Fe}_2\text{O}_3$) in all samples. A superparamagnetic (SPM) iron oxide phase ($\alpha\text{-Fe}_2\text{O}_3$ SPM) was measured in Fe_2O_3 particles smaller than 13.5 nm (table S4 and accompanying text). Iron was highly dispersed on CNF, $\gamma\text{-Al}_2\text{O}_3$, and SiO_2 , as evidenced by the presence of SPM nanoparticles exclusively. The iron oxide particles on $\alpha\text{-Al}_2\text{O}_3$ and $\beta\text{-SiC}$ had a broader size distribution, whereas the bulk Fe-Ti-Zn-K catalyst was primarily composed of large Fe_2O_3 particles.

The supported and bulk Fe catalysts were tested in the FT reaction at 1 bar and 350°C at

low CO conversion (0.5 to 1%) to restrict secondary hydrogenation of olefins (Table 1 and fig. S4). Catalytic activity is expressed as iron time yield (i.e., the number of CO moles converted to hydrocarbons per gram of iron per second). A high initial activity was observed for $\text{Fe}/\beta\text{-SiC}$ and Fe/CNF . The activity of Fe/CNF decreased continuously during the 15 hours of reaction; the activity of the $\text{Fe}/\beta\text{-SiC}$ catalyst increased during the first 5 hours of reaction, then decreased slowly afterward (fig. S4A). $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ exhibited a lower catalytic activity than Fe/CNF and $\text{Fe}/\beta\text{-SiC}$; however, it showed remarkable stability, as the activity remained constant over 15 hours. $\text{Fe}/\gamma\text{-Al}_2\text{O}_3$ and Fe/SiO_2 displayed a low catalytic activity, comparable to the bulk Fe catalysts (fig. S4B). The Fe-Cu-K- SiO_2 catalysts showed an initial activity approximately 4 times that of the Fe-Ti-Zn-K catalyst. Nevertheless, the iron time yield decreased rapidly to achieve comparable values after 15 hours of reaction.

One of the most important requirements for an FTO catalyst is to obtain the maximum production of the lower-olefins fraction while limiting methane selectivity to the lowest level possible. Fe/CNF and $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ exhibited high selectivity toward lower olefins ($\sim 60\%$ C) while directing comparatively little carbon to methane ($<25\%$ C)

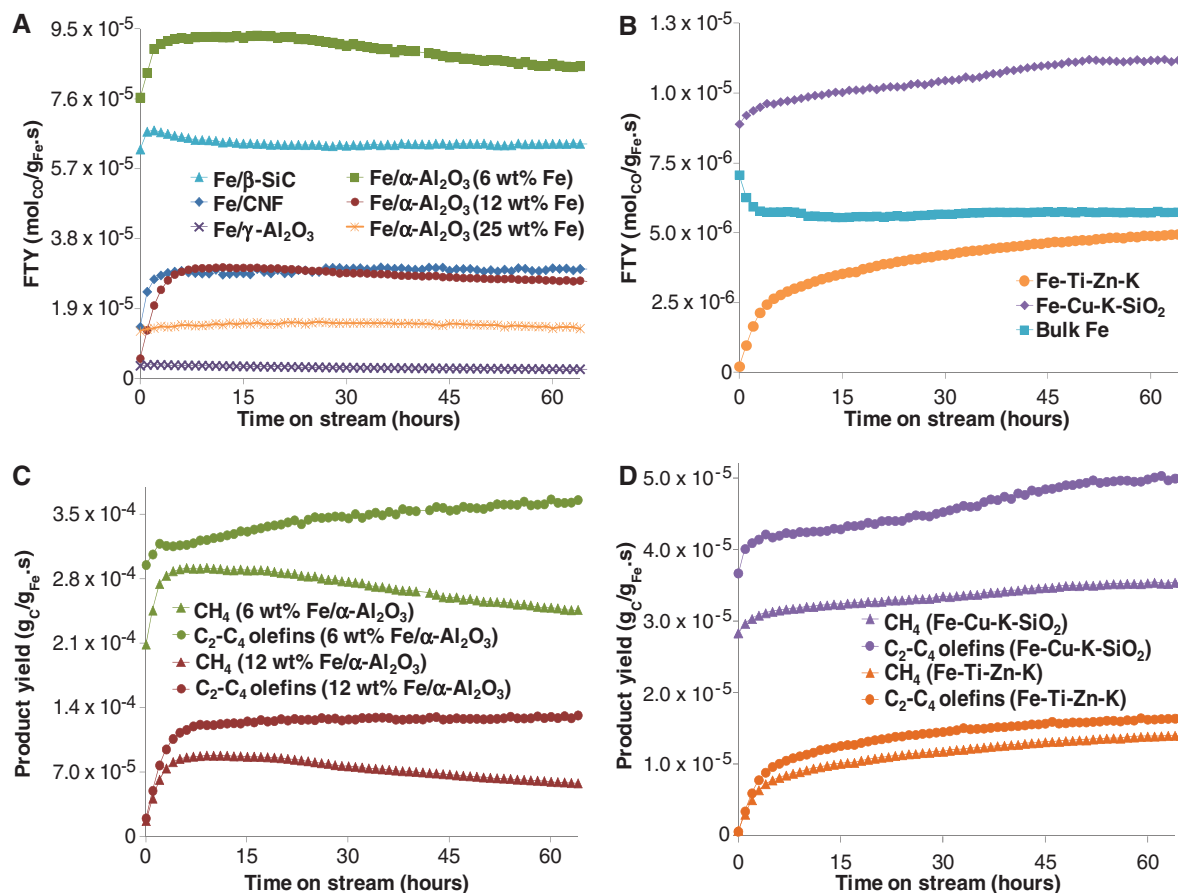


Fig. 2. (A to D) Catalytic performance of iron catalysts for the FTO process at 20 bar. Catalytic tests were carried out at $T = 340^\circ\text{C}$, $P = 20$ bar, and a H_2/CO ratio of 1. Iron time yield is plotted above as a function of time for (A) Fe-supported

catalysts and (B) bulk Fe catalysts. Methane and lower olefins yields are plotted below as a function of time for (C) Fe-supported catalysts and (D) bulk Fe catalysts. The product yields were obtained at CO conversion levels between 70 and 80%.

(Table 1). Fe/ β -SiC and Fe/SiO₂ also showed high selectivity to C₂ through C₄ olefins, but the CH₄ product fraction was higher than 30% C. Fe/ γ -Al₂O₃ and the bulk catalysts displayed a high selectivity to methane (>40% C), which is not desirable for their application in the FTO process.

Additional tests were carried out at 20 bar, 340°C, and an H₂/CO ratio of 1 to observe the per-

formance of supported and bulk Fe catalysts under industrially relevant conditions (Fig. 2). In view of the promising results obtained at 1 bar, we prepared and tested additional α -Al₂O₃-supported catalysts with different iron loadings (6 and 25 wt % Fe) to study the effect of iron content on catalytic performance. Most of the catalysts showed an initial increase in activity, except for the un-

promoted bulk Fe (Fig. 2B), which exhibited a decrease in activity during the first 10 hours of reaction before reaching stability. After an initial activation period, Fe/ β -SiC, Fe/CNF, 25 wt % Fe/ α -Al₂O₃, and Fe/ γ -Al₂O₃ showed a stable catalytic activity for 60 hours. A slight decrease in activity during reaction was observed for the 6 wt % and 12 wt % Fe/ α -Al₂O₃ catalysts, mainly resulting from a continuous drop in CH₄ production (Fig. 2C). The stability maintained during 60 hours fully complies with the requirements for the application of these catalysts in fluidized-bed reactors. In view of their favorable heat transfer characteristics, it is expected that these reactors will be preferred in industrial applications of the exothermic FTO process.

Table 2 summarizes the activities and product selectivities measured after 64 hours of reaction at 20 bar. The CO₂ selectivity for all the samples was approximately 40% on the basis of CO converted, except for Fe/ γ -Al₂O₃ (table S6). Under the selected reaction conditions, most of the catalysts had comparable CO conversion levels (77 to 81%; table S6). However, the Fe/ γ -Al₂O₃ catalyst only achieved a CO conversion of 10%. The promoted catalysts prepared using supports with low interaction with iron showed high catalytic activities combined with high selectivities to the desired products. Fe/CNF and 25 wt % Fe/ α -Al₂O₃ exhibited high selectivities toward C₂ through C₄ olefins (>50% C) while yielding a methane product fraction lower than 15% C. The Fe-Cu-K-SiO₂ catalyst showed a catalytic activity comparable to the 25 wt % Fe/ α -Al₂O₃; however, only moderate selectivities toward lower olefins were obtained.

The Anderson-Schulz-Flory model (eq. S2) that is used to predict the product distribution indicates that the maximum selectivity achievable for the C₂-C₄ fraction, including olefins and paraffins, is approximately 50 wt %, at a chain growth probability (α) between 0.4 and 0.5, as shown in fig. S5. This model predicts that methane selectivity is about 30 wt % when this maximum C₂-C₄ selectivity is reached.

Anderson-Schulz-Flory (ASF) plots (Fig. 3 and fig. S6) show that the catalysts prepared using inert supports provide α values of ~0.4, close to the optimal value for maximum lower-olefins production. Moreover, the plots in Fig. 3 revealed lower methane selectivities relative to the values predicted from the ASF model. This can be rationalized from the simplified “surface carbide” or “alkyl” mechanism (fig. S1), which is widely accepted for the FT synthesis (13). In this model, following CO dissociation and carbon hydrogenation, a CH₃ group adsorbed on the catalyst surface is proposed to act as a chain initiator. The carbon chain grows by the addition of methylene monomer units (CH₂) to the adsorbed alkyl species. The chain growth is terminated by β -hydride abstraction to form α -olefins or by hydrogenation to produce paraffins. Negative deviations from the ASF prediction for methane selectivity can be expected when using iron catalysts modified

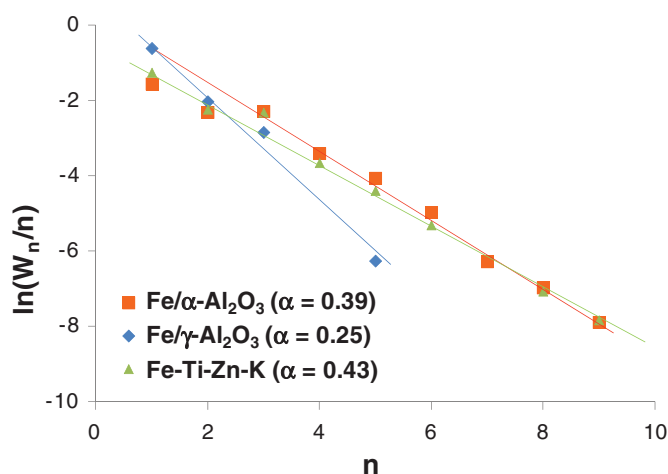
Table 1. Product selectivity and catalytic activity at 1 bar. Catalytic tests were performed at 350°C and a H₂/CO ratio of 1; results after 15 hours on stream are shown (CO conversion: 0.5 to 1.0%). The product mixture that was analyzed consisted of C₁ to C₁₆ hydrocarbons. Iron time yield (FTY) represents moles of CO converted to hydrocarbons per mol of Fe per second; %C is defined as carbon atoms in a product with respect to the total number of C atoms in the hydrocarbon mixture. CO₂ was not measured.

Sample	FTY (10 ⁻⁶ mol _{CO} /g _{Fe} ·s)	Selectivity (%C)			
		CH ₄	C ₂ -C ₄ olefins	C ₂ -C ₄ paraffins	C ₅ +
Fe/CNF	1.41	23	61	4	12
Fe/ α -Al ₂ O ₃ (12 wt % Fe)	0.65	22	61	4	13
Fe/ β -SiC	6.52	31	58	4	7
Fe/SiO ₂	0.14	38	56	5	1
Fe/ γ -Al ₂ O ₃	0.07	54	44	2	0
Fe-Ti-Zn-K	0.13	83	16	1	0
Fe-Cu-K-SiO ₂	0.20	43	46	2	9
Bulk Fe	0.08	76	21	2	1

Table 2. Catalytic performance at 20 bar. Catalytic tests were performed at 340°C and a H₂/CO ratio of 1; results after 64 hours on stream are shown. The product mixture that was analyzed consisted of C₁ to C₁₀ hydrocarbons. FTY and selectivity are defined as in Table 1. The selectivities were calculated on hydrocarbons produced; CO conversions and CO₂ selectivities are reported in table S6.

Sample	FTY (10 ⁻⁵ mol _{CO} /g _{Fe} ·s)	Selectivity (%C)				
		CH ₄	C ₂ -C ₄ olefins	C ₂ -C ₄ paraffins	C ₅ +	Oxygenates
Fe/CNF	2.98	13	52	12	18	5
Fe/ α -Al ₂ O ₃ (6 wt % Fe)	8.48	24	35	21	10	10
Fe/ α -Al ₂ O ₃ (12 wt % Fe)	2.66	17	39	19	14	11
Fe/ α -Al ₂ O ₃ (25 wt % Fe)	1.35	11	53	6	21	9
Fe/ β -SiC	6.38	35	19	39	4	3
Fe/ γ -Al ₂ O ₃	0.25	49	33	11	1	6
Fe-Ti-Zn-K	0.49	24	28	29	10	9
Fe-Cu-K-SiO ₂	1.12	26	36	12	18	8
Bulk Fe	0.57	30	32	18	14	6

Fig. 3. Comparison of ASF plots for supported and bulk catalysts. The ASF plots are based on the product distribution obtained when performing the FT reaction at 20 bar, 340°C, and a H₂/CO ratio of 1 after 64 hours time on stream. Fe/ α -Al₂O₃ has an iron loading of 12 wt %; n is the number of carbon atoms in a product, and W_n is the weight fraction of the product with carbon number equal to n .



with promoters that limit the hydrogenation reactions (8), thus favoring chain growth and the termination step via β -hydride abstraction that cannot give rise to CH_4 production. The suppression of the methanation reaction induced by the promoters was only observed when using CNF or $\alpha\text{-Al}_2\text{O}_3$ because these “inert” supports are thought to favor the proximity between iron and promoters (Na plus S), in contrast to reactive supports such as $\gamma\text{-Al}_2\text{O}_3$ that lead to more methane (Fig. 3). In the case of the bulk catalysts, CH_4 selectivities coincided with the values predicted by the ASF model or were slightly above.

Mössbauer spectroscopy of the spent catalysts after reaction at 1 bar (table S5) showed that the nature of the iron phases varied when using different support materials. Although some of the iron carbides may be oxidized after exposure to air, Fe_xC_y was detected on the samples with moderate to high catalytic activity. In contrast, the samples with the lowest catalytic activity, Fe/SiO_2 and $\text{Fe}/\gamma\text{-Al}_2\text{O}_3$, did not contain any carbides. A strong metal-support interaction clearly inhibits the formation of catalytically active iron carbides, as observed for conventional high-surface-area support materials. Note that in the size range of iron particles dispersed on inert supports (7 to 20 nm), particle size effects seem to be minimal.

TEM performed on spent catalysts revealed that the iron nanoparticles in the supported samples increased in size. The particle size distributions of the fresh and spent $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ and Fe/CNF are shown in fig. S2. For Fe/CNF , changes in the catalytic activity were only observed during the first 4 hours of reaction, which suggests that the changes in the catalyst structure took place during catalyst activation and initial usage. In the case of $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$, the average iron nanoparticle size increased from 14 ± 5 nm to 17 ± 5 nm (Fig. 1C). The promoted bulk iron oxide showed

extensive particle fragmentation and carbon filament growth, which brings about the poor mechanical stability of this catalyst (Fig. 1D).

The spent catalysts were characterized with thermogravimetric analysis to determine the extent of carbon lay-down. Carbon burn-off experiments were performed for all the samples, except for the Fe/CNF catalyst. Although extensive carbon deposition on the samples after reaction with CO-rich syngas and high temperatures could be expected, most of the samples exhibited low solid carbon formation. After 64 hours of reaction at 340°C and 20 bar, the levels of carbon lay-down measured on the spent catalysts were lower than 10 wt %. In contrast, $\text{Fe}/\alpha\text{-Al}_2\text{O}_3$ (25 wt % Fe) and Fe-Cu-K-SiO_2 exhibited a higher extent of coke formation (23 wt % and 40 wt %, respectively).

The FTO process represents a strong alternative route for the sustainable production of lower olefins from biomass-derived synthesis gas. The industrial potential of this process is greatly enhanced by the reported development of active, selective, and mechanically stable catalysts that consist of promoted iron nanoparticles dispersed on weakly interactive supports. Further suppression of methane production, maximization of the $\text{C}_2\text{-C}_4$ olefins fraction, and reduction of carbon lay-down by addition of promoters and by optimization of physical properties (e.g., Fe particle size, distribution of Fe nanoparticles on the support) will allow us to further understand and develop the performance of these catalysts.

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Supporting Online Material

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Materials and Methods
Eqs. S1 and S2
Tables S1 to S6
Figs. S1 to S6
References (27–35)

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Plate Motions and Stresses from Global Dynamic Models

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Delineating the driving forces behind plate motions is important for understanding the processes that have shaped Earth throughout its history. However, the accurate prediction of plate motions, boundary-zone deformation, rigidity, and stresses remains a difficult frontier in numerical modeling. We present a global dynamic model that produces a good fit to such parameters by accounting for lateral viscosity variations in the top 200 kilometers of Earth, together with forces associated with topography and lithosphere structure, as well as coupling with mantle flow. The relative importance of shallow structure versus deeper mantle flow varies over Earth's surface. Our model reveals where mantle flow contributes toward driving or resisting plate motions. Furthermore, subducted slabs need not act as strong stress guides to satisfy global observations of plate motions and stress.

Predicting plate motions correctly, along with stresses within the plates, has been a challenge for global dynamic models. Accurate

predictions are vitally important for understanding the forces responsible for the movement of plates, mountain building, rifting of continents,

and strain accumulation released in earthquakes. Previous studies have investigated these driving forces by either predicting stresses in the plates alone (1, 2) or plate motions alone (3–5). Other studies have taken the important step of predicting both plate motions and stresses in a single model (6–8). However, in addition to predicting plate motions, a successful global dynamic model must also explain plate rigidity and plate boundary-zone deformation, as well as intraplate stress patterns. Furthermore, the presence of lateral viscosity variations within the top 200 km of Earth influences the coupling between lithosphere and mantle convection. A systematic investigation of this influence is needed to improve our understanding of the driving mechanisms for plate tectonics.

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This PDF file includes:

Materials and Methods

Tables S1 to S6

Figs. S1 to S6

Eqs. S1 and S2

References

Materials and Methods

Catalyst preparation

Preparation of supported Fe catalysts. Fe supported catalysts with a nominal loading of 10 wt% Fe were prepared by multi-step incipient wetness impregnation of the selected support (2 g) with 2.7 ml of a 1.4M aqueous solution of ammonium iron citrate (90-100%, J. T. Baker). The ammonium iron citrate as purchased contained low amounts of sulphur and sodium. Different support materials were used: α -Al₂O₃ (10 m² g⁻¹; BASF), β -SiC (25 m² g⁻¹; SICAT), γ -Al₂O₃ (250 m² g⁻¹; BASF), SiO₂ (495 m² g⁻¹; Grace Davison) and CNF (150 m² g⁻¹). Before impregnation, the β -SiC support was calcined at 700°C under static air for 3 hours. The carbon nanofibers support was prepared and surface-oxidized as described before (27). The solution was added to the support until the pore volume was completely filled. After each impregnation step, the catalysts were dried at room temperature and 60 mbar for 2 hours. 6 wt% and 25 wt% Fe/ α -Al₂O₃ catalysts were prepared by impregnation of 4 g of support with 2.8 ml (1.3M) and 12.8 ml (1.4M) of aqueous solutions of ammonium iron citrate. Impregnation and drying steps were repeated until incorporation of the total volume of the solution was achieved. The dried impregnates, except for the CNF supported sample, were heated at 500°C for 2 h, at a heating rate of 10 K min⁻¹ under air flow. Impregnated CNF was heated at 355°C for 2 h, at a heating rate of 10 K min⁻¹ under nitrogen flow.

Preparation of bulk Fe catalysts. A bulk promoted iron oxide catalyst (Fe-Ti-ZnO-K₂O with a weight composition 100:25:10:4) was synthesized according to the procedure reported by Bössemeyer *et al.* (10). Additionally, two bulk (unpromoted Fe₂O₃ and Fe₂O₃/CuO/K₂O/SiO₂) catalysts were prepared by precipitation of an aqueous solution of ferric nitrate and adding a basic sodium carbonate solution as precipitating agent. 25 g of Fe(NO₃)₃·9H₂O (98% A.C.S. reagent, Acros) were dissolved in 100 ml of distilled water. For the promoted catalyst, 2.9 g of Cu(NO₃)₂·3H₂O (p.a. 99.5%, Merck) was added to the solution. The solution was heated to its boiling point and after that, a near boiling solution of 25 g of Na₂CO₃ (99%, Acros) in 100 ml of distilled water was slowly added while stirring vigorously. The precipitate was filtered and re-slurried in boiling water to remove residual sodium. The washing process was repeated until neutral pH was reached. The unpromoted catalyst was dried at 60°C for 6 h and then at 120°C for 24 h. Subsequently, the catalyst was calcined under air flow at 300°C for 5 h with a heating ramp of 5 K min⁻¹. For addition of K, after precipitation the Cu-promoted catalyst was re-slurried in 200 ml of distilled water. 16 g of potassium water glass solution (K₂O: SiO₂ 1:2.15, Akzo-PQ) were added to the slurry under vigorous stirring. 3 ml of HNO₃ was added to precipitate SiO₂ and to lower the potassium content. After this, the catalyst was washed, dried and calcined following the same procedure as for the unpromoted catalyst.

Catalyst characterization

X-ray fluorescence measurements were carried out to determine the iron loading and the elemental composition of the catalysts on Goffin Meyvis spectro X-lab 2000. X-ray diffraction measurements were performed with a Bruker AXS D8 ADVANCE diffractometer equipped with a CoK α 1 source (λ =0.1789 nm) in a wide angle range (from 20 to 90° in 2 θ). TEM images were obtained on a Philips Tecnai-20FEG (200 kV) microscope equipped with an EDX and HAADF detector. Thermal gravimetric analysis

of the spent catalysts to determine carbon lay-down was performed in a Perkin Elmer Pyris 1 TGA under oxygen flow (10 ml min⁻¹). Temperature was increased from 30°C to 600°C with a heating ramp of 5 °C min⁻¹. After reaching 600°C, the samples remained at that temperature for 1 hour. Transmission ⁵⁷Fe Mossbauer spectra were collected at room temperature with a conventional constant acceleration spectrometer using a ⁵⁷Co (Rh) source. Velocity calibration was carried out using an α -Fe foil. The Mössbauer parameters were determined by fitting the spectra with calculated subspectra consisting of Lorentzian-shaped lines, using the Mosswin 3.0i program.

Catalytic tests

Low pressure testing. Fischer-Tropsch synthesis was performed at 1 bar and 350°C in a plug flow reactor (H₂/CO ratio of 1). 20 mg of the catalyst (particle size = 0.2-0.4 mm) were diluted with 200 mg of SiC (particle size = 0.2 mm) to achieve isothermal plug-flow conditions and reduced *in situ* in H₂/Ar (33% v/v; 60 ml min⁻¹) at 350°C for 2 h (Heating ramp: 5 °C min⁻¹). Gas chromatography was used to determine the selectivity for hydrocarbons up to C₁₆. Catalytic activities and product selectivities were determined uptill 15 h of reaction at low CO conversions (0.5-1%). CO₂ selectivity was not measured. Product selectivity has been calculated as equivalent amount of carbon atoms in a product with respect to the total carbon atoms present in the hydrocarbons produced. The selectivity to oxygenates was below 1%C and has been excluded from the reported product selectivities.

Medium pressure testing. The Fischer-Tropsch reaction was carried out at 20 bar and 340°C in a plug flow reactor. 2 ml of catalyst (particle size = 0.2-0.4 mm) were diluted with 10 ml of SiC (particle size = 0.2-0.4 mm) and reduced *in situ* under H₂ flow (22 ml min⁻¹) at 350°C for 2 h. After reduction, temperature was lowered to 290°C and the feed flow was switched to a syngas/internal standard mixture (H₂/CO/He: 45/45/10%v/v, GHSV=1500 h⁻¹). Temperature was raised to 340°C (Heating ramp: 10 °C h⁻¹). Catalytic activities and product selectivities were determined up to 64 h of reaction. Product selectivity was determined by gas chromatography analysis of the products up to C₁₀. Analysis of the liquid product confirmed the production of hydrocarbons, mainly in the range of C₈ to C₁₅. Only trace amounts (<1%C) of alcohols (ethanol) were observed.

SOM Text

Mössbauer Spectroscopy

The Mössbauer spectral contributions consisting of quadrupole-split center lines are assigned to high-spin Fe³⁺ in an octahedral environment and, most likely, correspond to superparamagnetic Fe₂O₃ in the fresh samples. In magnetic nanocrystals with dimensions lower than 13.5 nm (34,35) the thermal energy is high enough to induce the random fluctuation of the magnetic moments, cancelling out the magnetic ordering. At low temperatures or in larger particles, the coupling forces prevail over the thermal energy, causing the magnetic moments of neighbouring atoms to align and the Mössbauer spectra becomes magnetically split (hyperfine sextuplets are observed).

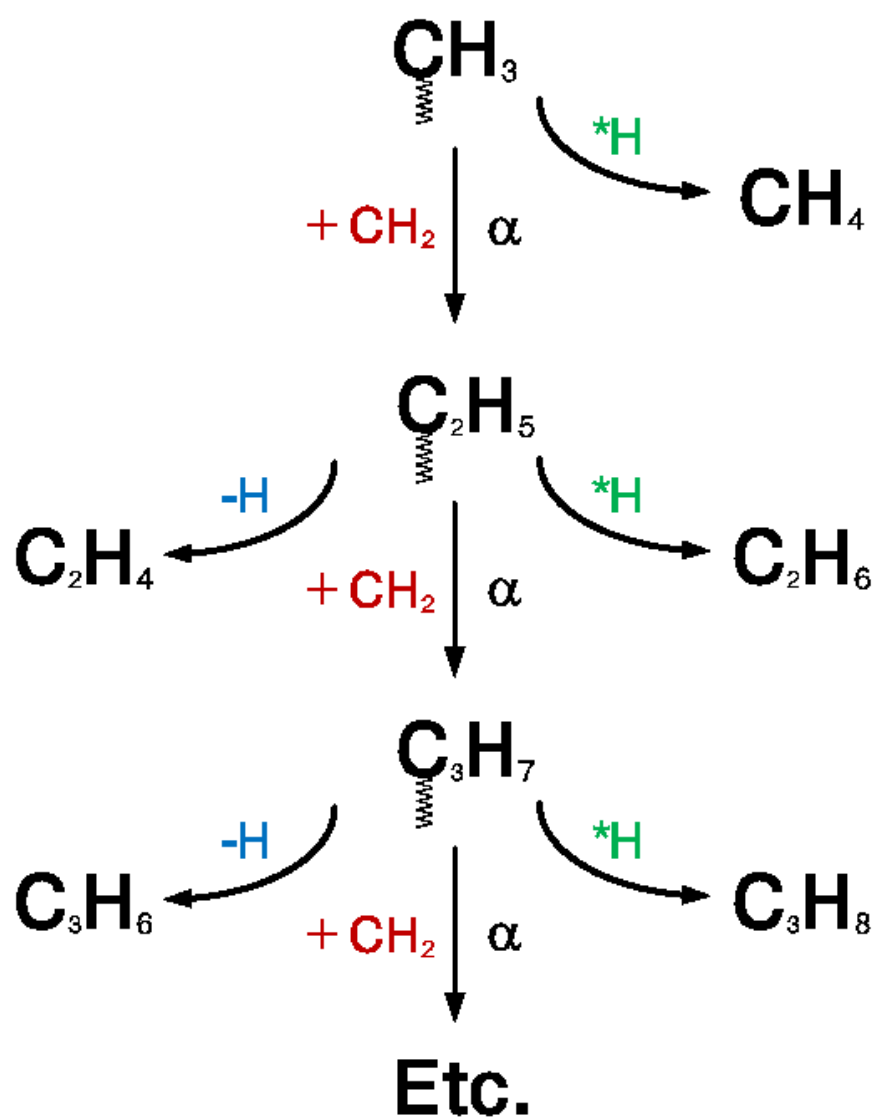


Fig. S1.

Fischer-Tropsch reaction mechanism (surface carbide mechanism). Methylene monomers (CH_2) are added to the adsorbed chain. The chain growth terminates by β -hydride elimination or by hydrogenation.

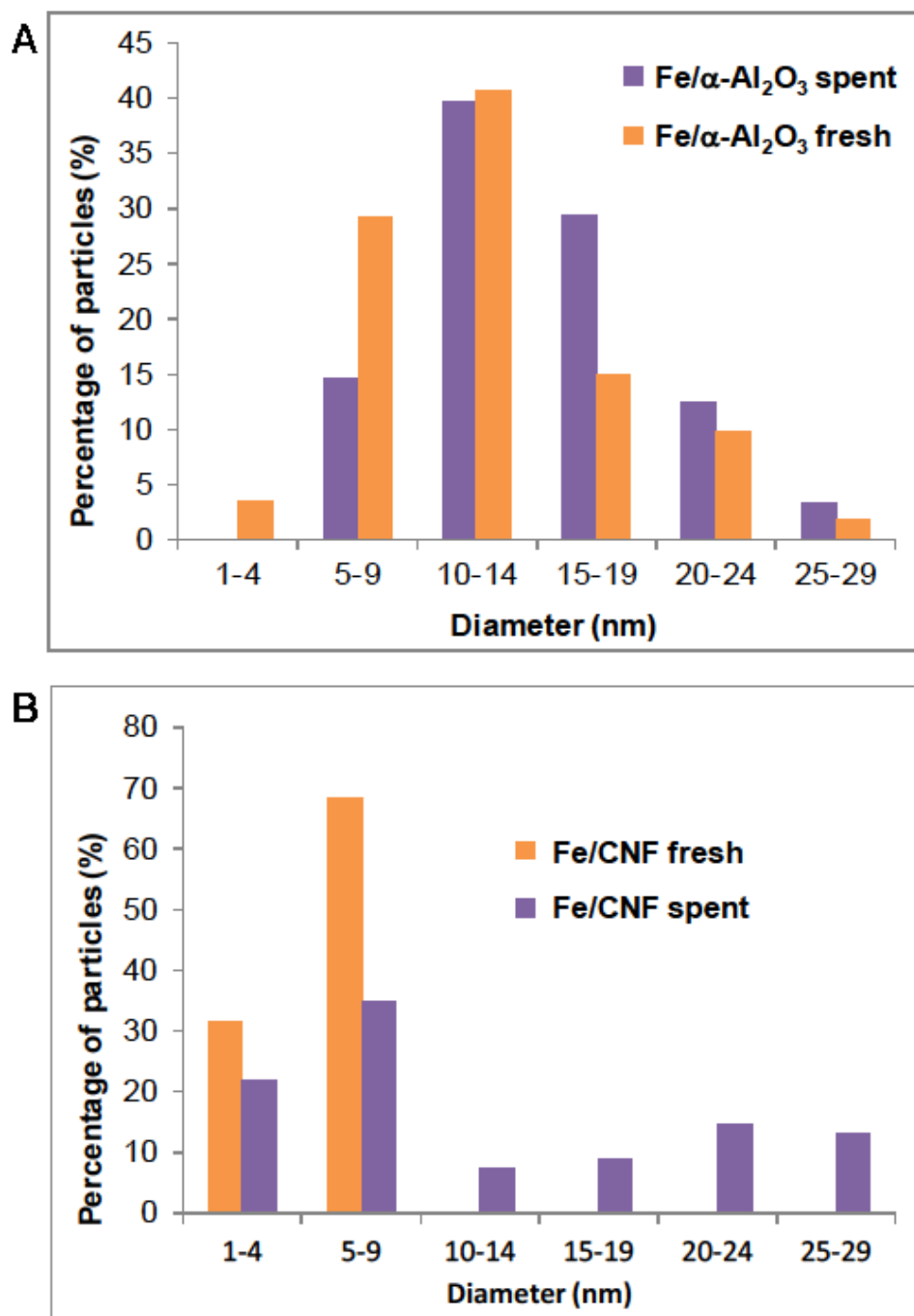


Fig. S2.

Particle size distribution of iron nanoparticles. (A) 12 wt% Fe/α-Al₂O₃, **(B)** Fe/CNF. Fresh catalysts: dark-coloured bars. The spent catalysts (light-coloured bars) were analyzed after reaction at 340°C, 20 bar and H₂/CO ratio of 1 (TOS = 64 h).

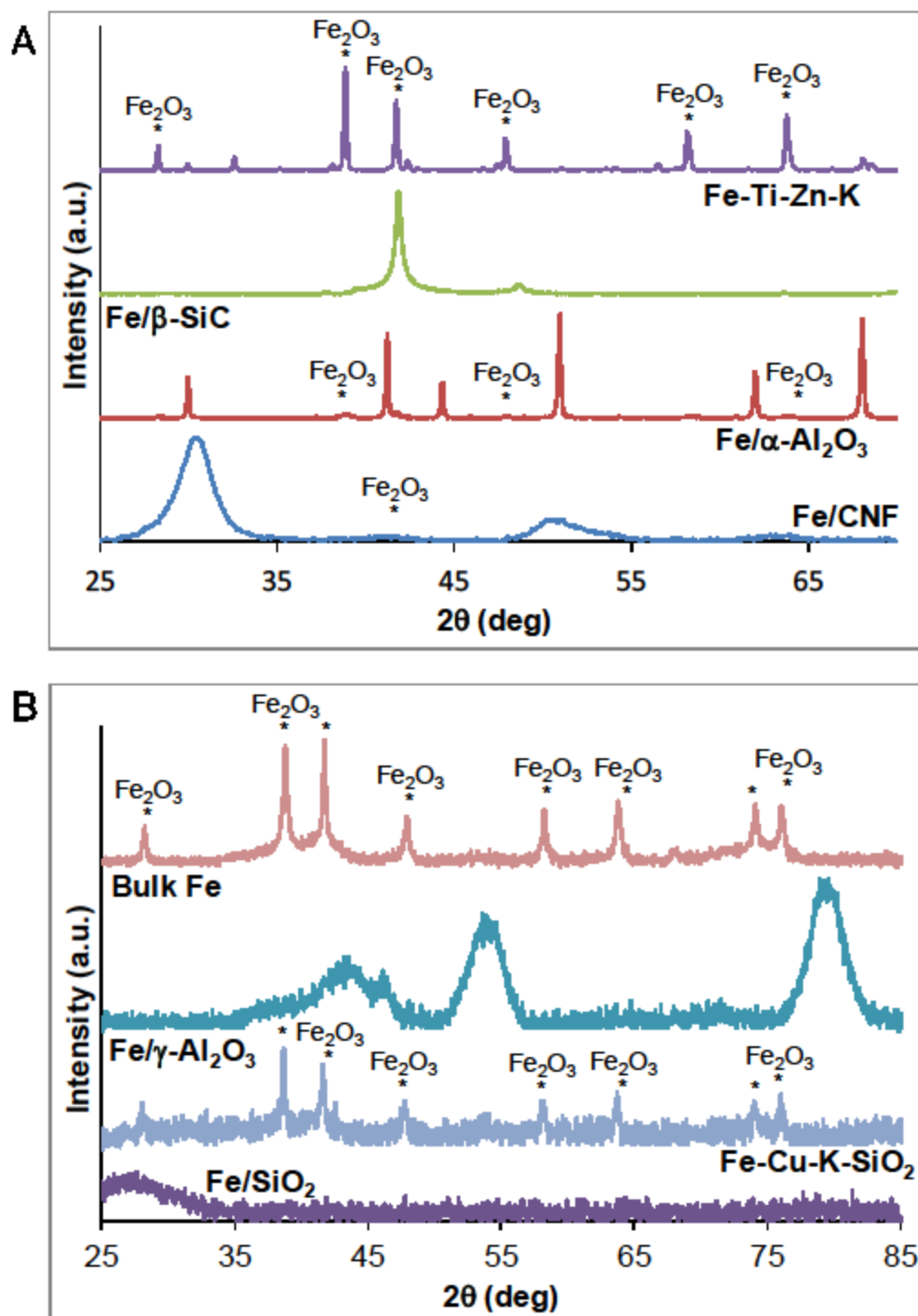


Fig. S3.

X-ray diffraction of supported and bulk Fe catalysts. Diffraction lines related to iron were not observed in catalysts supported on β -SiC (A), γ -Al₂O₃ and SiO₂ (B).

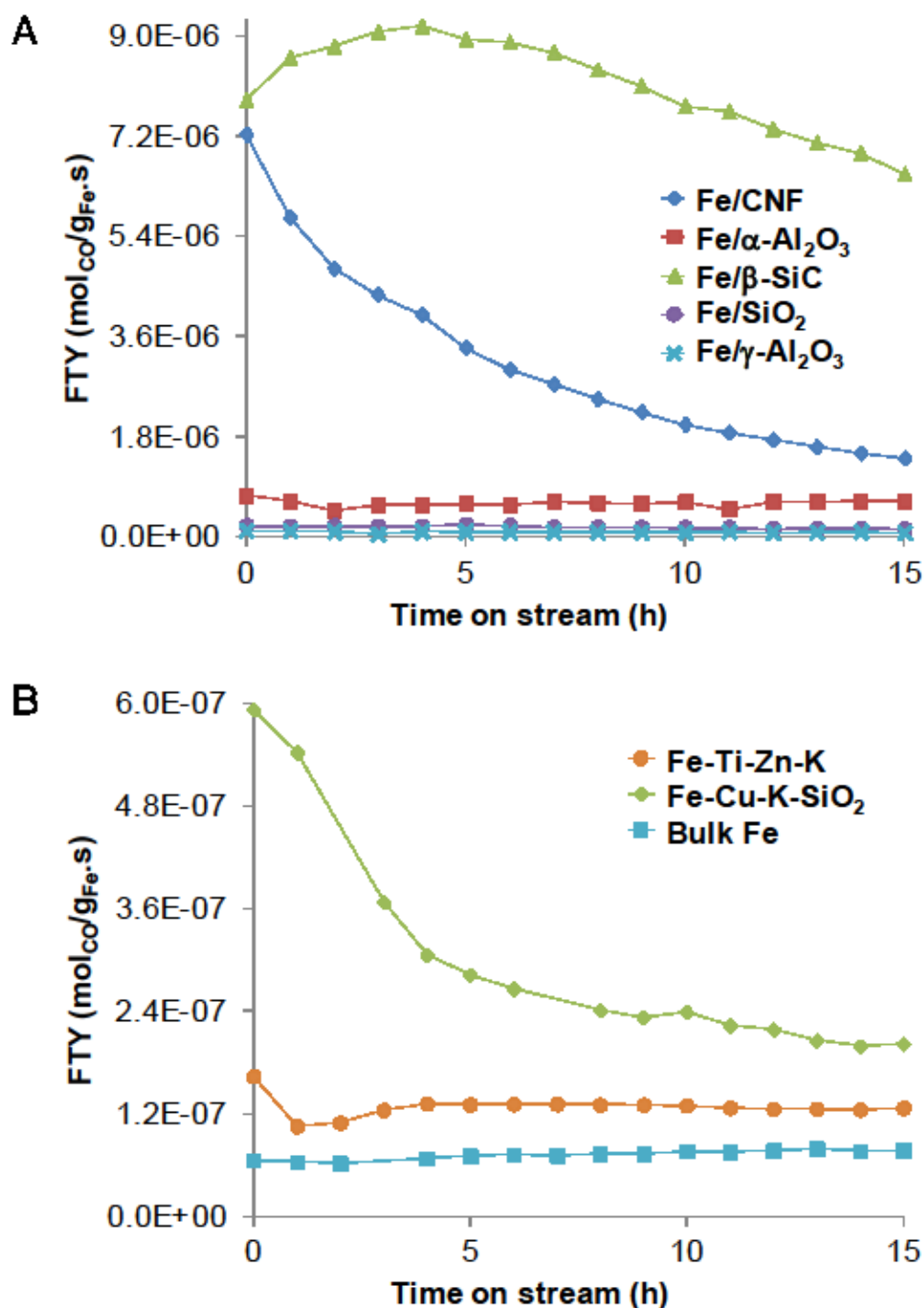


Fig. S4.
Catalytic activity as a function of time of iron catalysts for the Fischer-Tropsch to Olefins process at 1 bar. Catalytic activity was measured at T=350°C, P=1 bar, H₂/CO ratio of 1, and low CO conversion levels (0.5-1%). **(A)**, Iron time yield for supported Fe catalysts. **(B)**, Iron time yield as a function of time on stream (TOS) for Fe bulk catalysts.

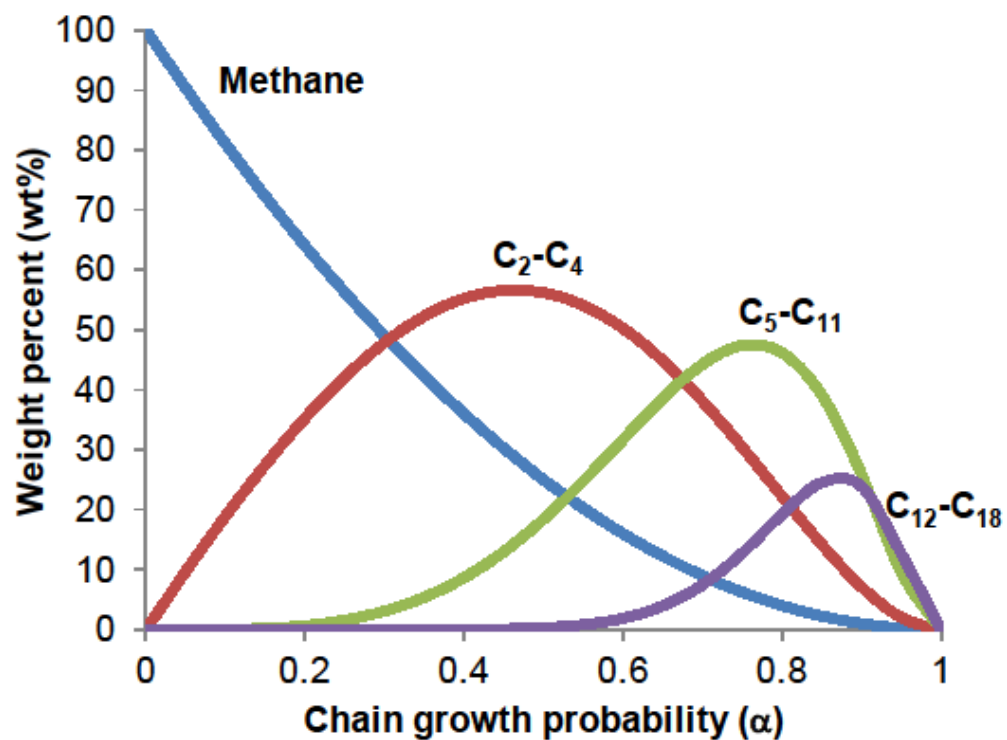


Fig. S5.
Product distribution predicted by the Anderson-Schulz-Flory model. The maximum C_2-C_4 fraction is obtained at α -values between 0.4 and 0.5.

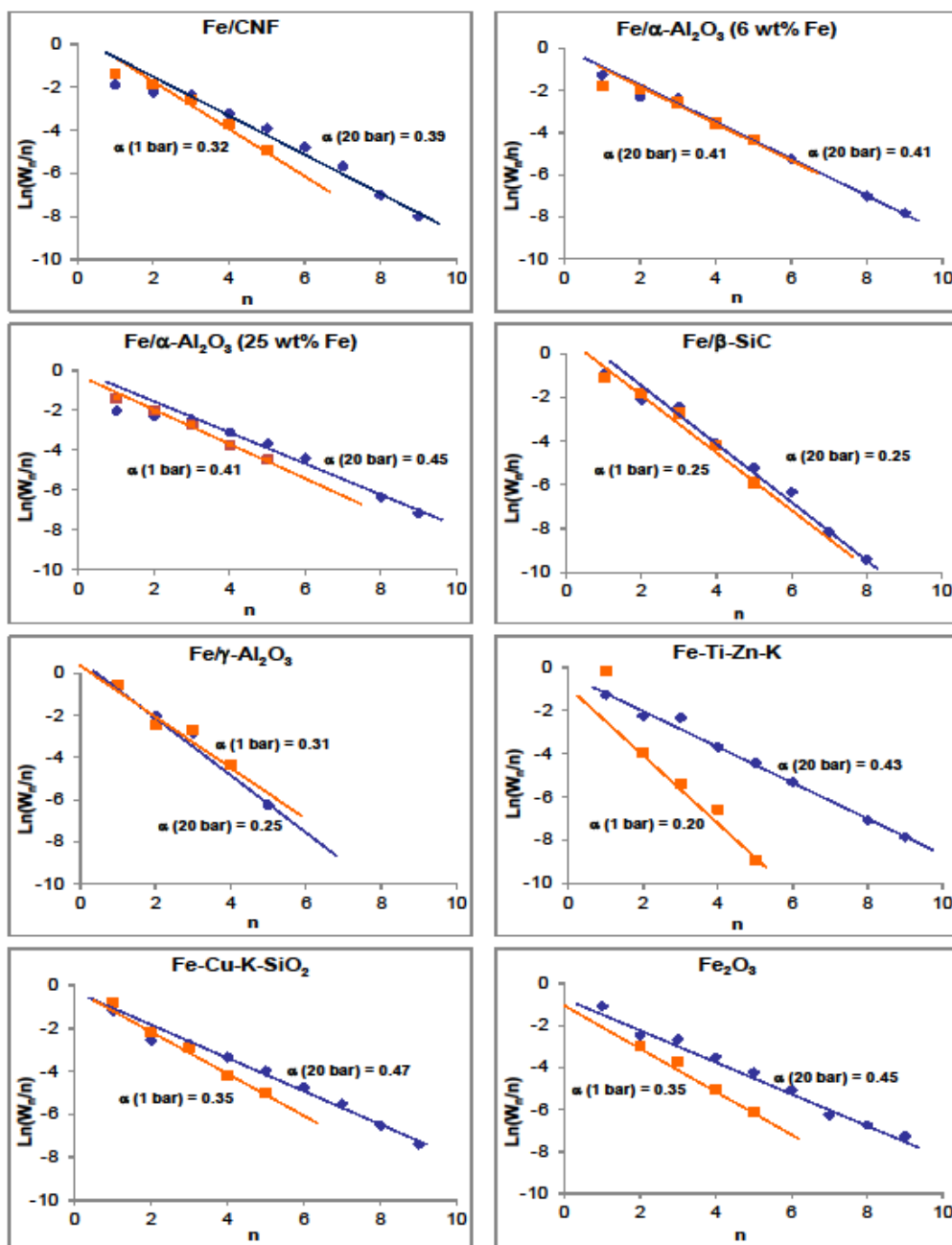


Fig. S6.

Anderson-Schulz-Flory plots. (■) Product distribution at 350°C, 1 bar and H_2/CO ratio of 1 (TOS = 15 h). (◆) Product distribution at 340°C, 20 bar and H_2/CO ratio of 1 (TOS = 64 h). W_n is the weight fraction of a product with n number of carbon atoms; α is the chain growth probability and $1-\alpha$ is the probability of chain termination.

Table S1.

Catalytic results from the literature on supported iron catalysts tested under high temperature conditions.

Catalyst	Fe (wt%)	P (bar)	T (°C)	H ₂ /CO	CO conv. (%)	FTY (mol _{CO} to HC/g _{Fe} .s)	Selectivity (%C)		
							CH ₄	C ₂ -C ₄ olefins	C ₅ +
Fe/Al ₂ O ₃ (15)	20	15	280	1	22	N. A.	26	9	35
Fe-Pr/HT γ -Al ₂ O ₃ (28)	2	8	280	0.5	5.4*	1.95x10 ⁻⁵	5	51	42
Fe-Pr/HT γ -Al ₂ O ₃ (TOS = 64 h)	2	20	350	1	18	4.3x10 ⁻⁵	42	23	8
Fe/AC (29)	4	10	275	1	2.24*	1.7x10 ⁻⁵ ‡	35	38 [¶]	9
Fe/AC (21)	4.4	1	350	1	2.3	1.6x10 ⁻⁴ ‡ 4.6x10 ⁻⁵ §	18	60 ^{†#}	5
Fe/CNT (30)	10.8	20	275	2	86	N. A.	9	13	70
Fe-Mn/C (31)	1.53	1	309	3	1.3*	N. A.	38	39	0
Fe-Ru-K/CNT (32)	9.8	8	275	2	25	1.0x10 ⁻⁴	15	N. A.	46
Fe-K/C spheres (33)	8.9	8	275	2	N.A.	1.8x10 ⁻⁵	3	4	92

* CO converted to hydrocarbons; † Selectivity towards C₂ and C₃ olefins; ‡ Initial activity; § Iron time yield after 20 h; N.A.= data not available.

^{||}This sample was prepared in our laboratory according to the procedure described by Baker *et al.*(28) and tested for comparison under our reaction conditions.

Please note that the activated carbon supported systems deactivated rapidly: ¶ the activity of this catalyst decreased 50% from its initial value after 17 h (H₂/CO ratio of 3) and # for this catalyst, activity decreased 71% after 20 h.

Table S2.Iron loading and average Fe₂O₃ crystallite size of fresh Fe catalysts.

Sample	Fe ₂ O ₃ crystallite size* (nm)	Fe loading [†] (wt%)
Fe/CNF	7	12
Fe/ α -Al ₂ O ₃	21	13
Fe/ β -SiC	N.A.	8
Fe/SiO ₂	N. A.	13
Fe/ γ -Al ₂ O ₃	N. A.	13
Fe-Ti-Zn-K	50	72
Fe-Cu-K-SiO ₂	40	32
Bulk Fe	30	63

N. A.: Not applicable

* Determined by X-ray diffraction (XRD).

[†] Determined by X-ray fluorescence (XRF).

Table S3.

Elemental analysis of the fresh catalysts determined with XRF (X-Ray Fluorescence).

Sample	Na (wt%)	Al (wt%)	Si (wt%)	Ca (wt%)	Sx (wt%)	Mn (wt%)	K (wt%)	Cr (wt%)
Fe/ α -Al ₂ O ₃ (12 wt%)	0.594	43.19	0.14	0.022	0.132	0.049	-	0.025
Fe/ β -SiC	0.245	0.091	41.42	0.028	0.045	0.036	-	0.015
Fe/SiO ₂	0.473	0.060	37.79	-	0.052	0.057	-	0.023
Fe/ γ -Al ₂ O ₃	0.496	42.28	0.137	-	0.062	0.057	-	0.020
Fe-Cu-K-SiO ₂	-	0.090	19.5	-	-	-	8.05	-
Bulk Fe	-	0.036	0.238	0.017	-	-	-	-

Table S4.

Mössbauer parameters of fresh Fe catalysts. Experimental uncertainties: Isomer shift: IS $\pm 0.01 \text{ mm.s}^{-1}$; Quadrupole splitting: QS $\pm 0.01 \text{ mm.s}^{-1}$; Line width: $\Gamma \pm 0.01 \text{ mm.s}^{-1}$; Hyperfine field: $\pm 0.1 \text{ T}$; Spectral contribution: $\pm 3\%$.

Sample	IS (mm.s^{-1})	QS (mm.s^{-1})	Γ (mm.s^{-1})	Hyperfine field (T)	Phase	Spectral Contribution (%)
Fe/CNF	0.36	0.81	0.58	-	$\alpha\text{-Fe}_2\text{O}_3$ (SPM)	100
Fe/ $\alpha\text{-Al}_2\text{O}_3$	0.34	0.81	0.83	-	$\alpha\text{-Fe}_2\text{O}_3$ (SPM)	24
	0.38	-0.21	0.46	50.8	$\alpha\text{-Fe}_2\text{O}_3$	76
Fe/ $\beta\text{-SiC}$	0.33	0.80	0.57	-	$\alpha\text{-Fe}_2\text{O}_3$ (SPM)	82
	0.39	-0.14	1.32	47.0*	$\alpha\text{-Fe}_2\text{O}_3$	18
Fe/SiO ₂	0.34	0.90	0.51	-	$\alpha\text{-Fe}_2\text{O}_3$ (SPM)	97
	0.39	-0.20	0.51	51.7	$\alpha\text{-Fe}_2\text{O}_3$	3
Fe/ $\gamma\text{-Al}_2\text{O}_3$	0.31	0.97	0.57	-	$\alpha\text{-Fe}_2\text{O}_3$ (SPM)	100
Fe-Ti-Zn-K	0.39	0.56	0.45	-	$\alpha\text{-Fe}_2\text{O}_3$ (SPM)	9
	0.38	-0.21	0.50	50.2	$\alpha\text{-Fe}_2\text{O}_3$	91
Fe-Cu-K-SiO ₂	0.33	0.73	0.49	-	$\alpha\text{-Fe}_2\text{O}_3$ (SPM)	87
	0.40	-0.21	0.47	51.8	$\alpha\text{-Fe}_2\text{O}_3$	13
Bulk Fe	0.33	0.72	0.51	-	$\alpha\text{-Fe}_2\text{O}_3$ (SPM)	67
	0.39	-0.23	0.48	51.6	$\alpha\text{-Fe}_2\text{O}_3$	33

* Mean magnetic hyperfine field.

Table S5.

Mössbauer parameters of spent Fe catalysts after 15 h of reaction at T=350°C, P=1 bar and H₂/CO ratio of 1. Experimental uncertainties: Isomer shift: IS $\pm$ 0.01 mm.s⁻¹; Quadrupole splitting: QS $\pm$ 0.01 mm.s⁻¹; Line width: Γ $\pm$ 0.01 mm.s⁻¹; Hyperfine field: $\pm$ 0.1 T; Spectral contribution: $\pm$ 3%.

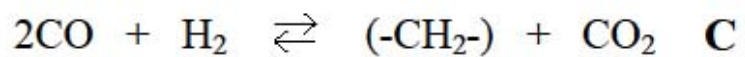
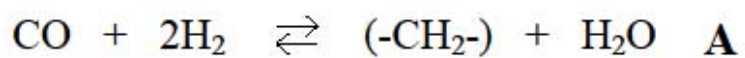
Sample	IS (mm.s ⁻¹)	QS (mm.s ⁻¹)	Γ (mm.s ⁻¹)	Hyperfine field (T)	Phase	Spectral Contribution (%)
Fe/CNF	0.21	-	0.71	16.4	χ -Fe ₅ C ₂ (I)	6
	0.28	-	0.71	20.3	χ -Fe ₅ C ₂ (II)	6
	0.23	-	0.71	10.1	χ -Fe ₅ C ₂ (III)	3
	0.31	0.74	0.71	-	Fe ³⁺	85
Fe/ α -Al ₂ O ₃	0.20	-	0.38	18.0	χ -Fe ₅ C ₂ (I)	31
	0.26	-	0.35	21.4	χ -Fe ₅ C ₂ (II)	28
	0.22	-	0.45	10.7	χ -Fe ₅ C ₂ (III)	20
	0.32	0.98	0.70	-	Fe ³⁺	21
Fe/ β -SiC	0.37	0.96	0.78	-	Fe ³⁺	66
	0.19	-	0.44	18.5	χ -Fe ₅ C ₂ (I)	14
	0.25	-	0.44	21.9	χ -Fe ₅ C ₂ (II)	13
	0.29	-	0.41	10.9	χ -Fe ₅ C ₂ (III)	7
Fe/SiO ₂	0.34	1.04	0.63	-	Fe ³⁺	100
Fe/ γ -Al ₂ O ₃	0.35	0.98	0.73	-	Fe ³⁺	90
	0.85	2.38	0.70	-	Fe ²⁺	10
Fe-Ti-Zn-K	0.23	-	0.77	18.1	χ -Fe ₅ C ₂ (I)	18
	0.28	-	0.72	21.7	χ -Fe ₅ C ₂ (II)	17
	0.24	-	0.41	11.5	χ -Fe ₅ C ₂ (III)	7
	0.44	0.50	0.55	-	Fe ³⁺	8
	0.70	2.00	0.53	-	Fe ²⁺	5
	0.30	-	0.58	46.9*	Fe ₃ O ₄ (I)	12
	0.68	-	0.54	44.1*	Fe ₃ O ₄ (II)	8
	0.57	-	2.01	36.6*	Fe ₃ O ₄ (II) [†]	25

* Mean magnetic hyperfine field; [†] Smear sextet.

Table S6.

Catalytic performance at 20 bar. Catalytic tests performed at 340°C and H₂/CO ratio of 1; results after 64 h on stream.

Sample	Total CO conversion (%)	CO ₂ (% of CO converted)	Chain growth probability α
Fe/CNF	88	42	0.39
Fe/ α -Al ₂ O ₃ (6 wt% Fe)	77	46	0.41
Fe/ α -Al ₂ O ₃ (12 wt% Fe)	81	41	0.39
Fe/ α -Al ₂ O ₃ (25 wt% Fe)	80	40	0.45
Fe/ β -SiC	77	42	0.25
Fe/ γ -Al ₂ O ₃	10	20	0.25
Fe-Ti-Zn-K	79	41	0.43
Fe-Cu-K-SiO ₂	79	37	0.47
Bulk Fe	97	34	0.45



Equation S1.

Reaction equations. **A**, General Fischer-Tropsch reaction. **B**, Water gas shift (WGS) reaction. **C**, Overall Fischer-Tropsch reaction valid on catalysts with WGS activity such as iron.

$$\ln \left(\frac{W_n}{n} \right) = n \ln \alpha + \ln \left[\frac{(1 - \alpha)^2}{\alpha} \right]$$

Equation S2.

Anderson-Schulz-Flory model for the product distribution. n is the number of carbon atoms in a product; W_n is the weight fraction of a product with n number of carbon atoms; α is the chain growth probability and $1-\alpha$ is the probability of chain termination.

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